

Time Response of feedback control systems

Control systems are generally called upon to perform both under transient and steady conditions. A feedback control system has the inherent capability that its parameters can be adjusted to alter both transient and steady state behaviour. In order to analyse the transient and steady state behaviour of control systems, the first step always is to obtain a mathematical model of the system. For any specific input signal a complete time response expression can then be obtained through the Laplace Transform inversion.

Time Response: The response given by the system which is function of the time, to the applied excitation is called time response of a control system.

Transient Response: The output variation during the time, it takes to achieve its final value is called as transient response.

Steady state Response: It is that part of the time response which remains after complete transient response vanishes from the system output.

It can also be defined as response of the system as time approaches infinity from the time at which transient response completely dies out.

Standard Test Signals:

From the analysis point of view, the signals which are most commonly used as reference inputs are defined as standard test signals. The standard test signals are,

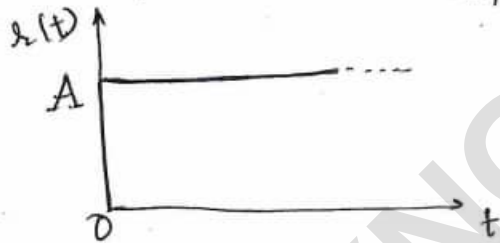
- ① Step signal: The step is a signal whose value changes from one level (usually zero) to another level 'A' in zero time. The mathematical representation of the step function is

$$x(t) = A \cdot u(t)$$

where

$$u(t) = \begin{cases} 1 & ; t > 0 \\ 0 & ; t < 0 \end{cases}$$

In the Laplace transform form, $R(s) = A/s$
The graphical representation of a ^{step} signal is as shown below,

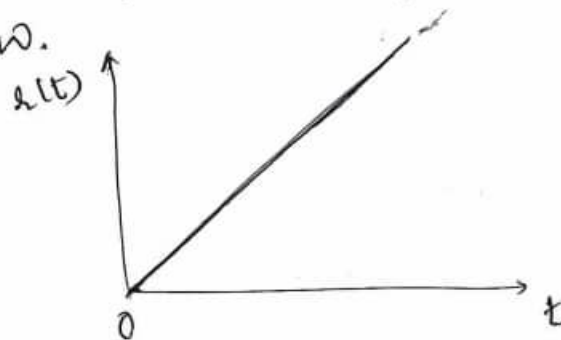


- ② Ramp signal: The ramp is a signal which starts at a value of zero and increases linearly with time. Mathematically,

$$x(t) = \begin{cases} At & ; t > 0 \\ 0 & ; t < 0 \end{cases}$$

In the Laplace transform form, $R(s) = A/s^2$

The graphical representation of a ramp signal is as shown below.



From the defining eq^s of step signal and ramp signal, it is seen that a ramp signal is integral of step signal.

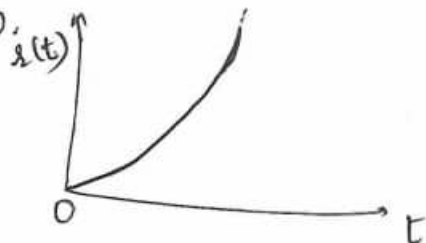
(3)

③ Parabolic signal: The mathematical representation of this signal is,

$$x(t) = \begin{cases} A \frac{t^2}{2} & ; t > 0 \\ 0 & ; t < 0 \end{cases}$$

In the Laplace transform form, $R(s) = A/s^3$

The graphical representation of a parabolic signal is shown below.



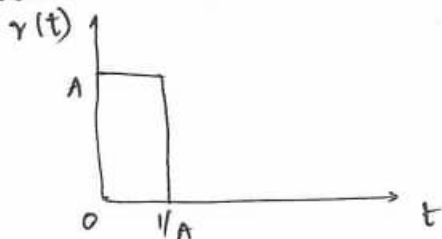
From the defining eqns of ramp signal and parabolic signal, it is seen that a parabolic signal is integral of a ramp signal.

④ Impulse signal: A unit impulse is defined as a signal which has zero value everywhere except at $t=0$, where its magnitude is infinite. It is generally called the δ -function and has the following property.

$$\delta(t) = 0 ; t \neq 0$$

$$\text{and } \int_{-\epsilon}^{+\epsilon} \delta(t) dt = 1 \quad \text{where } \epsilon \text{ tends to zero.}$$

Since a perfect impulse cannot be achieved in practice, it is usually approximated by a pulse of small width but unit area as shown below.

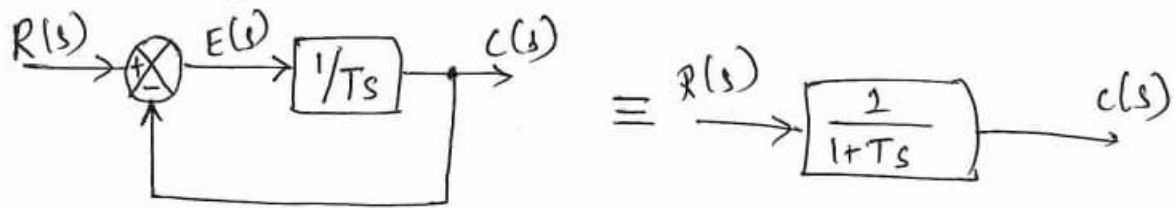


Mathematically, an impulse function is the derivative of a step signal i.e. $\delta(t) = \dot{u}(t)$

The Laplace transform of a unit impulse is, $R(s) = 1$

Time Response of First-Order systems:

Let us consider the first order system with unity feedback



Block diagram of a first-order system

The transfer function of the system of the above block diagram is,

$$\frac{C(s)}{R(s)} = \frac{1}{1+Ts}$$

Unit-step Response of the First Order Systems:

For the unit-step input $R(s) = \frac{1}{s}$, the output response $C(s)$ is given by,

$$C(s) = \frac{1}{s(1+Ts)}$$

Using Partial fraction expansion,

$$\frac{1}{s(1+sT)} = \frac{A}{s} + \frac{B}{1+sT}$$

$$1 = A(1+sT) + B \cdot s \rightarrow (1)$$

Put $s = 0$, in eqⁿ (1)

$$1 = A + 0 \therefore \boxed{A = 1}$$

Put $s = -\frac{1}{T}$ in eqⁿ (1)

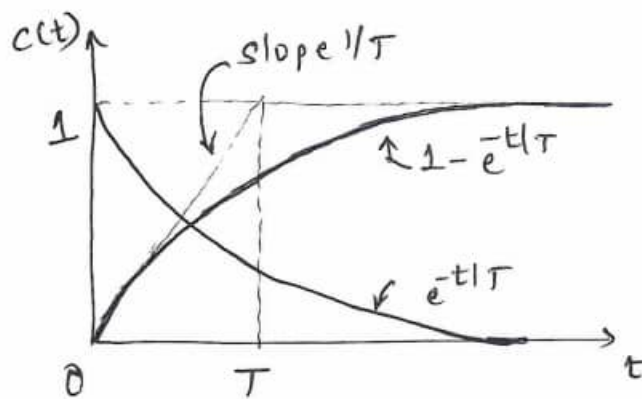
$$1 = 0 - \frac{B}{T} \therefore \boxed{B = -T}$$

$$\therefore C(s) = \frac{1}{s} - \frac{T}{1+sT}$$

Taking the inverse Laplace Transform, we get

$$\boxed{c(t) = 1 - e^{-t/T}}$$

which is plotted below,



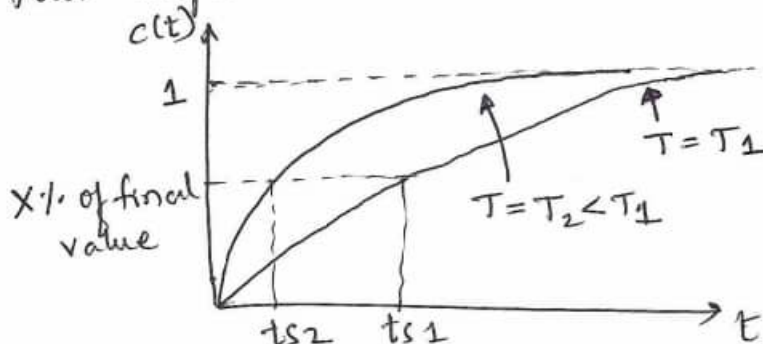
It is seen that the output rises exponentially from zero value to the final value of unity. The initial slope of curve at $t=0$ is given by,

$$\left. \frac{dc}{dt} \right|_{t=0} = \left. \frac{1}{T} e^{-t/T} \right|_{t=0} = \frac{1}{T}$$

where T is known as the time constant of the system. At this slope the output reaches final value $c=1$ in one time constant as shown in the above figure.

The time constant is indicative of how fast the system tends to reach the final value. The speed of response can be quantitatively defined as the time for the output to become a particular percentage of its final value.

A large time constant corresponds to a sluggish system and a small time constant corresponds to a fast response as shown below.



Effect of time constant on system response

Consider now, the error response of the system which is given by, $e(t) = r(t) - c(t) = e^{-t/T}$

The steady-state error is given by,

$$C_{ss} = \lim_{t \rightarrow \infty} e(t) = 0$$

Thus this system tracks the unit-step input with zero steady-state error.

Time Response of Second-order systems:

Every practical system takes finite time to reach to its steady state and during this period, it oscillates or increases exponentially.

Every system has a tendency to oppose the oscillatory behaviour of the system which is called damping.

This damping is measured by a factor or a ratio called damping ratio of the system. This factor explains how much dominant the opposition from the system is to the oscillations in the output.

Damping ratio is denoted by a Greek symbol Zeta ξ . As this measures the opposition by the system to the oscillatory behaviour, if it is made zero, ($\xi=0$) system will oscillate with maximum frequency. As there is no opposition from system, it oscillates naturally and freely. Hence this frequency of oscillations under $\xi=0$ condition is called natural frequency of oscillations of the system and is denoted by ω_n rad/sec.

The closed-loop transfer function for a standard second order system takes the form as,

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

The characteristic eqⁿ can be written as,

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

Unit step Response of Second-order systems:

For the unit step input $R(s) = \frac{1}{s}$, the output response $C(s)$ is given by,

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\xi\omega_n s + \omega_n^2)} \rightarrow (1)$$

Finding the roots of the characteristic eqⁿ:

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

$$\text{i.e. } s_{1,2} = \frac{-2\xi\omega_n \pm \sqrt{4\xi^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$s_{1,2} = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

The roots of the characteristic eqⁿ are given by,

$$\begin{aligned} s^2 + 2\xi\omega_n s + \omega_n^2 &= (s - s_1)(s - s_2) \\ &= (s + \xi\omega_n + \omega_n\sqrt{\xi^2 - 1})(s + \xi\omega_n - \omega_n\sqrt{\xi^2 - 1}) \end{aligned}$$

For $\xi > 1$; the roots are real, unequal and negative and such systems are 'Overdamped'.

For $\xi = 1$; the roots are real, equal and negative and such systems are 'Critically damped'.

For $\xi < 1$; the roots are complex conjugates with negative real part and such systems are 'Underdamped'.

Consider underdamped condition i.e. $\xi < 1$

The roots are $s_{1,2} = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$

as $\xi < 1$, $s_{1,2} = -\xi\omega_n \pm j\omega_n\sqrt{1 - \xi^2}$

(Since $\xi < 1$, the term $\sqrt{\xi^2 - 1}$ is written as $j\sqrt{1 - \xi^2}$)

Substituting these in eqⁿ ① we get,

$$C(s) = \frac{\omega_n^2}{s(s + \xi\omega_n + j\omega_n\sqrt{1 - \xi^2})(s + \xi\omega_n - j\omega_n\sqrt{1 - \xi^2})} \rightarrow \text{②}$$

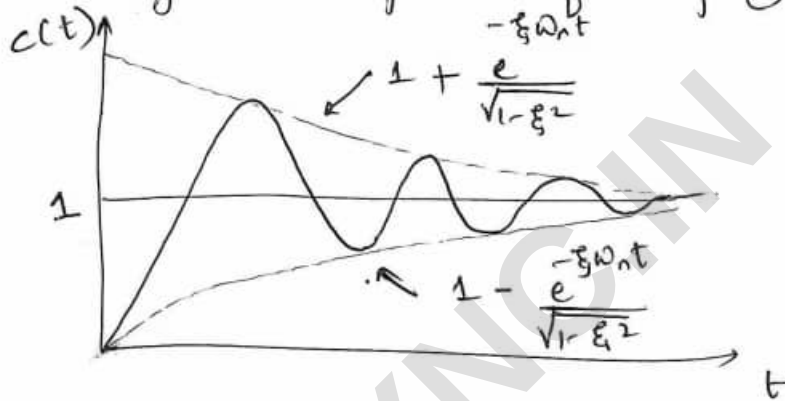
The Laplace inverse of the above eqⁿ, by the method of residues, can be obtained as,

$$c(t) = 1 - \frac{e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \sin \left[\omega_n \sqrt{1-\xi^2} \cdot t + \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi} \right]; t > 0$$

The steady-state value of $c(t)$ is given by,

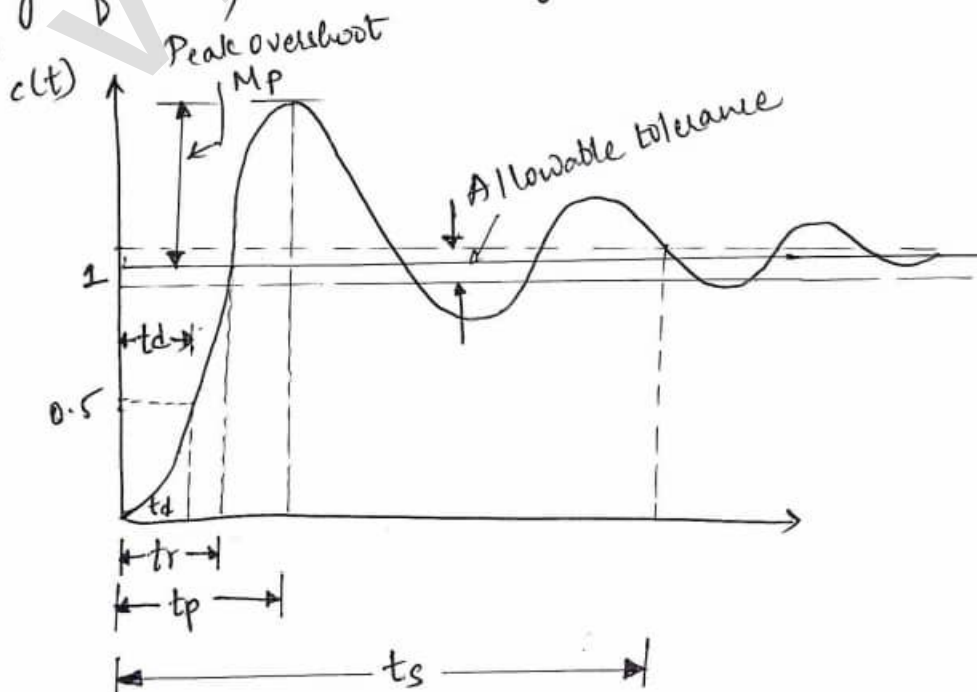
$$C_{ss} = \lim_{t \rightarrow \infty} c(t) = 1$$

The time response of an underdamped ($\xi < 1$) second-order system is plotted from eqⁿ (2) as,



Time Response Specifications :

The time response of second and higher-order control systems to a step input is generally of damped oscillatory nature as shown below.



Time Response Specifications

The various time response specifications are,

- ① Delay time (t_d): It is the time required for the response to reach 50% of the final value in first attempt.
- ② Rise time (t_r): It is the time required for the response to rise from 10% to 90% of the final value for overdamped systems and 0 to 100% of the final value for underdamped systems.
- ③ Peak time (t_p): It is the time required for the response to reach the peak of time response or the peak overshoot.
- ④ Peak overshoot (M_p): It indicates the normalized difference between the time response peak and the steady output and is defined as

$$\text{Peak percent overshoot} = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\%$$

- ⑤ Settling time (t_s): It is the time required for the response to reach and stay within a specified tolerance band (usually 2% or 5%) of its final value.
- ⑥ Steady-state error (e_{ss}): It indicates the error between the actual output and desired output as 't' tends to infinity, i.e.,

$$e_{ss} = \lim_{t \rightarrow \infty} [r(t) - c(t)]$$

Time Response Specifications of Second-Order Systems:

- ① Rise time (t_r): Time required by the output to achieve 100% of its final value, starting from zero during the first attempt is the rise time.

$$[c(t)]_{t=t_r} = 1$$

$$\text{i.e. } 1 - \frac{e^{-\xi \omega_n t r}}{\sqrt{1-\xi^2}} \sin \left[\omega_n \sqrt{1-\xi^2} t r + \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi} \right] = 1$$

$$- \frac{e^{-\xi \omega_n t r}}{\sqrt{1-\xi^2}} \sin \left[\omega_n \sqrt{1-\xi^2} t r + \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi} \right] = 0$$

The eqⁿ will get satisfied only if,

$$\sin \left[\omega_n \sqrt{1-\xi^2} t r + \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi} \right] = 0$$

Trigonometrically this is true only if,

$$\omega_n \sqrt{1-\xi^2} t r + \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi} = n\pi \quad \text{where } n=1, 2, 3, \dots$$

As we are interested in first attempt, let $n=1$

$$\therefore \omega_n \sqrt{1-\xi^2} t r + \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi} = \pi$$

Solving for 'tr' we get,

$$\therefore tr = \frac{\pi - \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n \sqrt{1-\xi^2}} \text{ sec}$$

② Peak time (t_p):

Transient response of second order underdamped system is given by,

$$c(t) = 1 - \frac{e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta) \quad \text{where } \omega_d = \omega_n \sqrt{1-\xi^2}$$

$$\text{and } \theta = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}$$

As at $t=t_p$, $c(t)$ will achieve its maxima.

According to maxima theorem,

$$\left. \frac{d}{dt} c(t) \right|_{t=t_p} = 0$$

$\therefore$ Differentiating $c(t)$ and equating it to zero,

$$0 - (-\xi \omega_n) \frac{e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta) - \frac{e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \omega_d \cos(\omega_d t + \theta) = 0 \quad (11)$$

Substituting $\omega_d = \omega_n \sqrt{1-\xi^2}$ in the second part of eq²,

$$\frac{\xi \omega_n e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta) - \frac{e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \omega_n \sqrt{1-\xi^2} \cos(\omega_d t + \theta) = 0$$

$$\therefore \frac{\omega_n e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \left\{ \xi \sin(\omega_d t + \theta) - \sqrt{1-\xi^2} \cos(\omega_d t + \theta) \right\} = 0$$

$$\therefore \xi \sin(\omega_d t + \theta) - \sqrt{1-\xi^2} \cos(\omega_d t + \theta) = 0$$

Dividing the eq² by $\cos(\omega_d t + \theta)$ we get,

$$\xi \tan(\omega_d t + \theta) - \sqrt{1-\xi^2} = 0$$

$$\therefore \tan(\omega_d t + \theta) = \frac{\sqrt{1-\xi^2}}{\xi}$$

But we know that $\theta = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi} \therefore \tan \theta = \frac{\sqrt{1-\xi^2}}{\xi}$

From the above two eq^s we can write,

$$\tan(\omega_d t + \theta) = \tan \theta$$

from Trigonometry we can write,

$$\tan(n\pi + \theta) = \tan \theta$$

$$\therefore \omega_d t = n\pi \quad \text{where } n = 1, 2, 3, \dots$$

As $t = t_p$, time required for first peak overshoot.

$$\therefore n = 1$$

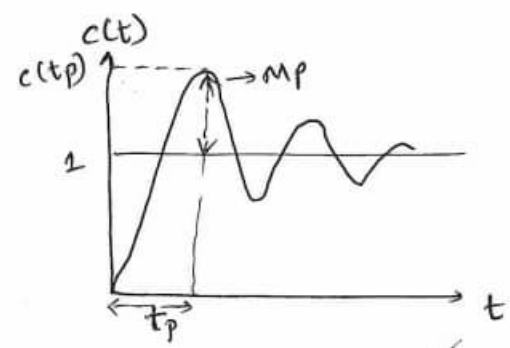
$$\therefore \omega_d t_p = \pi$$

$$\therefore t_p = \frac{\pi}{\omega_d}$$

$$\therefore \boxed{t_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}}$$

③ Peak overshoot (M_p):

Considering the diagram, the eqⁿ for M_p can be written as,



$$M_p = c(tp) - 1$$

$$M_p = \left\{ 1 - \frac{e^{-\xi \omega_n t_p}}{\sqrt{1-\xi^2}} \sin(\omega_d t_p + \theta) \right\} - 1$$

where $\omega_d = \omega_n \sqrt{1-\xi^2}$
and $\theta = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}$

$$\therefore M_p = -\frac{e^{-\xi \omega_n t_p}}{\sqrt{1-\xi^2}} \sin(\omega_d t_p + \theta)$$

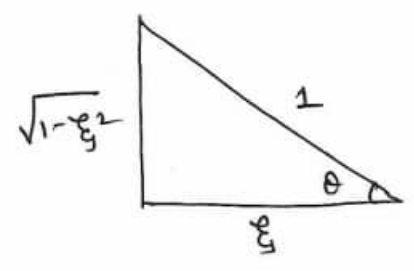
We know that, $t_p = \frac{\pi}{\omega_d}$, substituting this in the above eqⁿ,

$$M_p = -\frac{e^{-\xi \omega_n t_p}}{\sqrt{1-\xi^2}} \sin(\pi + \theta)$$

$$\therefore M_p = \frac{e^{-\xi \omega_n t_p}}{\sqrt{1-\xi^2}} \sin \theta$$

$\because \sin(\pi + \theta) = -\sin \theta$

We can write $\tan \theta = \frac{\sqrt{1-\xi^2}}{\xi}$



$$\sin \theta = \frac{\sqrt{1-\xi^2}}{1} \quad \text{[Opp/Hyp]}$$

$$\therefore M_p = \frac{e^{-\xi \omega_n t_p}}{\sqrt{1-\xi^2}} \cdot \frac{\sqrt{1-\xi^2}}{1}$$

$$M_p = \frac{e^{-\xi \omega_n t_p}}{1}$$

Substituting for $t_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}$

$$\therefore M_p = \frac{e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}}}{1}$$

$$\therefore \% M_p = 100 e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}}$$

④ Settling Time (t_s): It is the time required for the response to reach and stay within a specified tolerance band (usually 2% or 5%) of its final value.

$$\therefore c(t) \Big|_{t=t_s} = 0.98 \quad \text{For 2\%}$$

At $t=t_s$, the transient oscillatory term completely vanishes. The only term which controls the amplitude of the output within 2% is $e^{-\xi\omega_n t}$.

$$\therefore c(t_s) = 1 - e^{-\xi\omega_n t_s}$$

$$\text{i.e. } 0.98 = 1 - e^{-\xi\omega_n t_s}$$

$$\frac{e^{-\xi\omega_n t_s}}{e} = 0.02$$

$$-\xi\omega_n t_s = \ln(0.02)$$

$$-\xi\omega_n t_s = -3.912$$

$$\therefore t_s = \frac{3.912}{\xi\omega_n}$$

In practice the settling time is assumed to be,

$$t_s = \frac{4}{\xi\omega_n}$$

where $\frac{1}{\xi\omega_n}$ is called 'time constant' T

$$\therefore t_s = 4T$$

For 2% Tolerance

Similarly for 5% of tolerance band,

$$c(t_s) = 0.95$$

$$\text{i.e. } 0.95 = 1 - e^{-\xi\omega_n t_s}$$

$$\therefore t_s = \frac{3}{\xi\omega_n}$$

$$\therefore t_s = 3T$$

for 5% Tolerance

Numericals :

(14)

- ① A second order system is given by $\frac{C(s)}{R(s)} = \frac{25}{s^2 + 6s + 25}$
find its rise time, peak time, peak overshoot and settling time if subjected to unit step input. Also calculate expression for its output response.

Solⁿ: Comparing the transfer functions with the standard form $\frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$

$$\omega_n^2 = 25 \quad \text{and} \quad 2\xi\omega_n = 6$$

$$\omega_n = 5$$

$$\therefore \xi = 0.6$$

$$\text{Rise time, } t_r = \frac{\pi - \tan^{-1} \sqrt{1 - \xi^2}}{\xi \omega_n \sqrt{1 - \xi^2}}$$

$$= \frac{\pi - 0.9272}{4}$$

$$\therefore t_r = 0.5535 \text{ sec}$$

$$\text{Peak time, } t_p = \frac{\pi}{\omega_n \sqrt{1 - \xi^2}} = 0.785 \text{ sec}$$

$$\text{Peak overshoot, } \%M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} \times 100$$

$$M_p = 9.48\%$$

$$\text{Settling time, } t_s = \frac{4}{\xi\omega_n} = 1.33 \text{ sec}$$

The expression for the output response will be,

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1 - \xi^2}} \sin(4t + 0.9272)$$

$$c(t) = 1 - 1.5625 e^{-3t} \sin(4t + 0.9272)$$

- (2) A system is given by differential equation,
 $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 8y = 8x$, where y = output and x = input.
 Determine all time domain specifications for unit step input.

Solⁿ: To find the transfer function $\frac{Y(s)}{X(s)}$, take the Laplace transform of above eqⁿ,

$$s^2 Y(s) + 4s Y(s) + 8Y(s) = 8X(s)$$

$$(s^2 + 4s + 8) Y(s) = 8X(s)$$

$$\therefore \frac{Y(s)}{X(s)} = \frac{8}{s^2 + 4s + 8} \text{ comparing with } \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\therefore \omega_n^2 = 8 \quad 2\xi\omega_n = 4$$

$$\omega_n = 2.83$$

$$\xi = 0.7067$$

$$\text{Rise time } t_r, \quad t_r = \frac{\pi - \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n \sqrt{1-\xi^2}}$$

$$= \frac{\pi - 0.7853}{2.83 \sqrt{1-0.5}} = \frac{2.356}{2.001}$$

$$\therefore t_r = 1.176 \text{ sec}$$

$$\text{Peak time, } t_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}} = \frac{\pi}{2.002}$$

$$\therefore t_p = 1.57 \text{ sec}$$

$$\text{Peak overshoot, } \%M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} \times 100$$

$$\therefore M_p = 4.33\%$$

$$\text{Settling time, } t_s = \frac{4}{\xi\omega_n} = \frac{4}{0.7067 \times 2.83} = \underline{\underline{2 \text{ sec}}}$$

$$\therefore C(t) = 1 - 1.41 e^{-2t} \sin(2t + \pi/4)$$

- ③ A signal is represented by the equation $\frac{d^2\theta}{dt^2} + 10\frac{d\theta}{dt} = 150e$ where $e = (r - \theta)$ is the actuating signal. Calculate the value of damping ratio, undamped and damped frequency of oscillation. Also draw the block diagram of the system, and find its closed loop transfer function.

Solⁿ:

Using $e = r - \theta$ in the eqⁿ:

$$\frac{d^2\theta}{dt^2} + 10\frac{d\theta}{dt} = 150r - 150\theta$$

$$\frac{d^2\theta}{dt^2} + 10\frac{d\theta}{dt} + 150\theta = 150r$$

Taking the Laplace transform,

$$s^2\theta(s) + 10s\theta(s) + 150\theta(s) = 150R(s)$$

$$[s^2 + 10s + 150]\theta(s) = 150R(s)$$

$$\therefore \frac{\theta(s)}{R(s)} = \frac{150}{s^2 + 10s + 150}$$

Comparing this with $\frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$

$$\omega_n^2 = 150 \quad \text{and} \quad 2\xi\omega_n = 10$$

$$\omega_n = \sqrt{150} \quad \xi = 0.4082$$

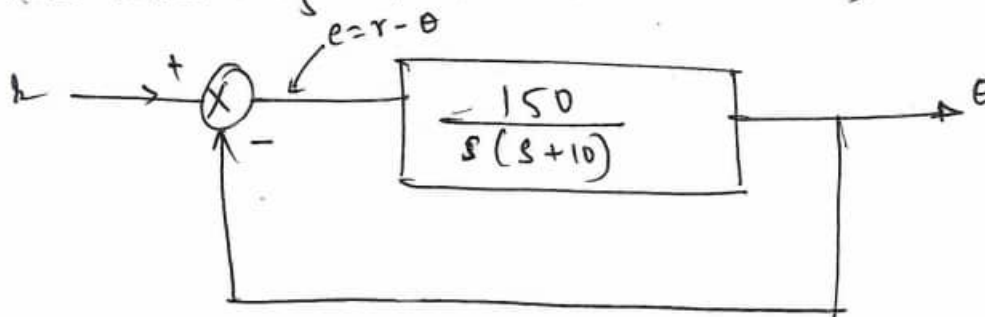
$$= 12.24$$

Damping ratio, $\xi = 0.4082$

Undamped frequency of oscillation $\omega_n = 12.24 \text{ rad/sec}$

Damped frequency of oscillation $\omega_d = \omega_n\sqrt{1-\xi^2} = 11.1806 \text{ rad/sec}$

The block diagram is as shown below,

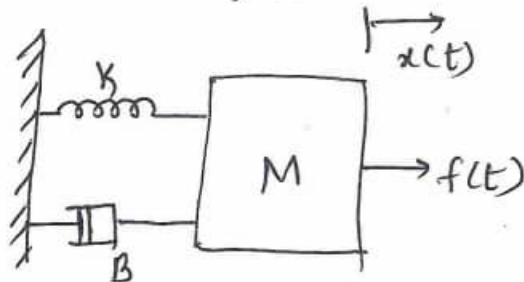


④ Referring to the figure, find the following.

(i) Transfer function $\frac{X(s)}{F(s)}$

(ii) $\xi, \omega_n, \%Mp, T_s$ and T_p

Take: $K = 33 \text{ N/m}$, $B = 15 \text{ N-s/m}$, $M = 3 \text{ kg}$



Solⁿ: (i) The differential eqⁿ can be written as,

$$f(t) = M \frac{d^2 x(t)}{dt^2} + B \frac{dx(t)}{dt} + K x(t)$$

Taking Laplace Transform on both sides, we get,

$$F(s) = M s^2 X(s) + B s X(s) + K X(s)$$

$$\therefore \frac{X(s)}{F(s)} = \frac{1}{M s^2 + B s + K}$$

$$= \frac{1/M}{s^2 + \frac{B}{M} s + \frac{K}{M}}$$

(ii) for $K=33$, $B=15$ and $M=3$

$$\frac{X(s)}{F(s)} = \frac{1/3}{s^2 + 5s + 11}$$

Comparing the denominator with $s^2 + 2\xi\omega_n s + \omega_n^2$

$$\omega_n^2 = 11 \quad \therefore \omega_n = 3.3166 \text{ rad/sec}$$

$$2\xi\omega_n = 5 \quad \therefore \xi = 0.7537$$

$$\therefore \%Mp = \frac{e^{-\pi\xi/\sqrt{1-\xi^2}}}{e} \times 100 = 2.7234\%$$

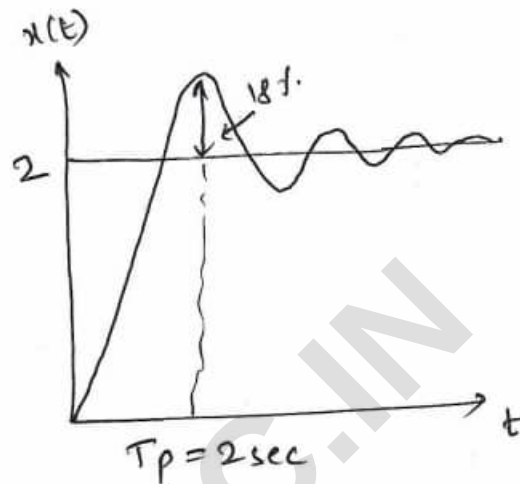
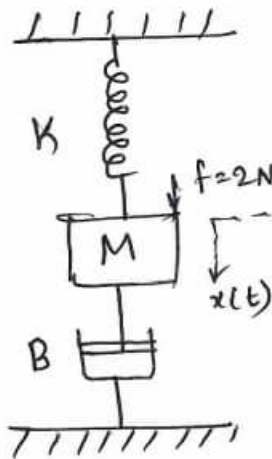
$$\omega_d = \omega_n \sqrt{1-\xi^2} = 2.1797 \text{ rad/sec}$$

$$\therefore T_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}} = 1.4412 \text{ sec}$$

$$\therefore T_s = \frac{4}{\xi\omega_n} = 1.6 \text{ sec}$$

- ⑤ For a spring mass damper system shown below, an experiment was conducted by applying a force of 2N to the mass. The response $x(t)$ was recorded using an xy plotter and the experimental result is as shown below. Find the value of M , K and B .

Solⁿ:



Solⁿ:

$$\therefore f(t) = M \frac{d^2 x}{dt^2} + B \frac{dx}{dt} + Kx$$

Taking Laplace Transform,

$$F(s) = [Ms^2 + Bs + K] X(s)$$

$$\therefore \frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Bs + K} = \frac{1/M}{s^2 + \frac{B}{M}s + \frac{K}{M}} \rightarrow (1)$$

Comparing the denominator with $s^2 + 2\xi\omega_n s + \omega_n^2$

$$\omega_n^2 = \frac{K}{M}, \quad \omega_n = \sqrt{\frac{K}{M}} \rightarrow (2) \quad \text{and} \quad 2\xi\omega_n = \frac{B}{M}, \quad \xi = \frac{B}{2\sqrt{KM}} \rightarrow (3)$$

Given $f(t) = 2$, $F(s) = \frac{2}{s}$ using in eqⁿ (1)

$$\therefore X(s) = \frac{2}{s} \left[\frac{1/M}{s^2 + \frac{B}{M}s + \frac{K}{M}} \right]$$

From the output, the steady state value of $x(t)$ is $X_{ss} = 2$

$$\therefore X_{ss} = \lim_{s \rightarrow 0} s X(s) = \lim_{s \rightarrow 0} \left[\frac{s^2 \left(\frac{1/M}{s^2 + \frac{B}{M}s + \frac{K}{M}} \right)}{s} \right] \Rightarrow 2 = \frac{2}{K}$$

$$\boxed{\therefore K = 1}$$

Given, the % overshoot is 18%.

$$\therefore M_p = 0.18$$

$$\therefore 0.18 = e^{\frac{-\pi \xi}{\sqrt{1-\xi^2}}} \Rightarrow \ln(0.18) = \frac{-\pi \xi}{\sqrt{1-\xi^2}}$$

$$\therefore -9.07147 = \frac{-\pi \xi}{\sqrt{1-\xi^2}} \Rightarrow (1-\xi^2) = 3.3564 \xi^2$$

$$\therefore \underline{\xi = 0.4791}$$

$$\text{And } T_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}} = 2$$

$$\therefore \omega_n = \frac{\pi}{2 \sqrt{1-\xi^2}} \Rightarrow \frac{\pi}{2 \sqrt{1-(0.4791)^2}}$$

$$\therefore \underline{\omega_n = 1.7895 \text{ rad/sec}}$$

Using eqⁿ (3) and (4)

$$\omega_n = \sqrt{K/M} \Rightarrow 1.7895 = \sqrt{1/M}$$

$$\boxed{\therefore M = 0.3122}$$

$$\xi = \frac{B}{2\sqrt{KM}} \Rightarrow 0.4791 = \frac{B}{2\sqrt{1 \times 0.3122}}$$

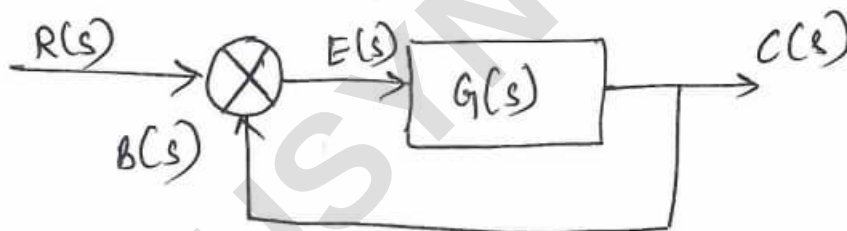
$$\boxed{\therefore B = 0.53544}$$

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Steady state errors and error constants:

Steady state errors constitute an extremely important aspect of system performance, for it would be meaningless to design for dynamic accuracy if the steady state output differed substantially from the desired value for one reason or another. The steady-state error is a measure of system accuracy. These errors arise from the nature of inputs, type of system and from nonlinearities of system components such as static friction, backlash etc. These are generally aggravated by amplifier drifts, aging or deterioration.

Consider a unity feedback system shown below.



Let the input be $R(s)$, the output be $C(s)$, the feedback signal be $B(s)$ and the difference between input and output is the error signal $E(s)$.
From the above figure, we see that,

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

And the output $C(s)$ can be written as,

$$C(s) = E(s) G(s)$$

$$\therefore E(s) = \frac{C(s)}{G(s)}$$

Substituting for $C(s)$ we get,

$$E(s) = \frac{\frac{G(s)}{1+G(s)} R(s)}{G(s)}$$

$$\therefore E(s) = \frac{1}{1+G(s)} \cdot R(s)$$

The steady state error e_{ss} may now be found by use of the final value theorem as follows.

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s)$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1+G(s)}$$

The above is the expression for steady-state error. The expression for steady-state errors for various types of standard test signals are derived below.

① Unit-step input :

$$\text{Input } x(t) = u(t)$$

$$\text{Laplace transform, } R(s) = \frac{1}{s}$$

We know that the steady state error,

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1+G(s)} \quad \text{as } R(s) = 1/s$$

$$= \frac{1}{1+G(0)} = \frac{1}{1+K_p}$$

where $K_p = G(0)$ is defined as Position error constant.

② Unit-ramp input :

$$\text{Input } x(t) = t$$

$$\text{Laplace transform, } R(s) = \frac{1}{s^2}$$

We know that the steady state error,

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot R(s)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} \quad \text{as } R(s) = \frac{1}{s^2}$$

$$= \lim_{s \rightarrow 0} \frac{1}{sG(s)} = \frac{1}{K_v}$$

where $K_v = \lim_{s \rightarrow 0} sG(s)$ is defined as Velocity error constant.

(3) Unit-parabolic input:

$$\text{Input } h(t) = \frac{t^2}{2}$$

$$\text{Laplace transform } R(s) = \frac{1}{s^3}$$

We know that the steady state error,

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot R(s)}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2G(s)} \quad \text{as } R(s) = 1/s^3$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2G(s)} = \frac{1}{K_a}$$

where $K_a = \lim_{s \rightarrow 0} s^2G(s)$ is defined as Acceleration error constant.

Types of Feedback Control System:

The open loop transfer function of a unity feedback system can be written in two standard forms: The time constant form and The pole-zero form.

In these two forms, $G(s)$ is as given below.

$$G(s) = \frac{K(T_{z1}s + 1)(T_{z2}s + 1) \cdots}{s^n(T_{p1}s + 1)(T_{p2}s + 1) \cdots} \quad \text{(Time constant form)} \rightarrow (1)$$

$$G(s) = \frac{K'(s + z_1)(s + z_2) \cdots}{s^n(s + p_1)(s + p_2) \cdots} \quad \text{(Pole-zero form)}$$

Eq² ① involves the term s^n in the denominator which corresponds to number of integrations in the system. As ' s ' tends to zero, this term dominates in determining the steady state error. Control systems are therefore classified in accordance with the number of integrations in the open loop transfer function $G(s)$ as described below.

① Type-0 system:

If $n=0$, the steady state errors to various standard inputs, obtained earlier, are

$$e_{ss}(\text{position}) = \frac{1}{1+G(0)} = \frac{1}{1+K} \neq \frac{1}{1+K_p}$$

$$e_{ss}(\text{Velocity}) = \lim_{s \rightarrow 0} \frac{1}{sG(s)} = \infty$$

$$e_{ss}(\text{acceleration}) = \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)} = \infty$$

Thus a system with $n=0$ or no integration in $G(s)$ has a constant position error, infinite velocity and infinite acceleration errors.

② Type-1 system:

If $n=1$, the steady-state errors to various standard inputs are,

$$e_{ss}(\text{position}) = \frac{1}{1+G(0)} = \frac{1}{1+\infty} = 0 \quad (\text{Referring to eq}^2 \text{ ①})$$

$$e_{ss}(\text{velocity}) = \lim_{s \rightarrow 0} \frac{1}{sG(s)} = \frac{1}{K} = \frac{1}{K_v}$$

$$e_{ss}(\text{acceleration}) = \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)} = \infty$$

Thus a system $n=1$ or with one integration in $G(s)$ has zero position error, a constant velocity error and an infinite acceleration error at steady-state.

③ Type-2 system :

If $n=2$, the steady-state errors to various standard inputs are,

$$e_{ss}(\text{position}) = \frac{1}{1+G(0)} = \frac{1}{1+\infty} = 0$$

$$e_{ss}(\text{velocity}) = \lim_{s \rightarrow 0} \frac{1}{sG(s)} = 0$$

$$e_{ss}(\text{acceleration}) = \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)} = \frac{1}{K} = \frac{1}{K_a}$$

Thus a system with $n=2$ or two integrations in $G(s)$ has a zero position error, zero velocity error and a constant acceleration error at steady-state.

Steady-state errors for various inputs and systems are summarized below.

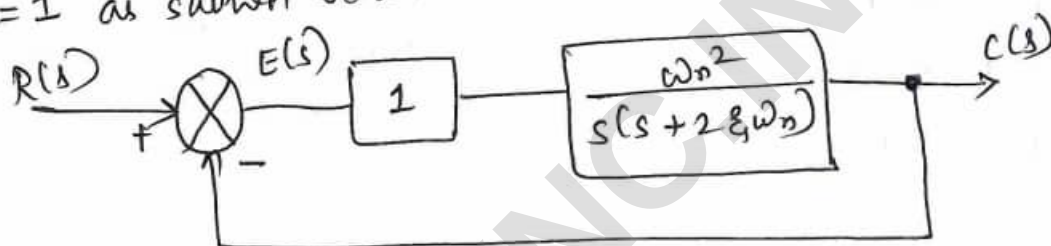
Type of Input	Steady-state error		
	Type-0 system	Type-1 system	Type-2 system
Unit-step	$1/(1+K_p)$	0	0
Unit-ramp	∞	$1/K_v$	0
Unit-parabolic	∞	∞	$1/K_a$
	where, $K_p = \lim_{s \rightarrow 0} G(s)$	where, $K_v = \lim_{s \rightarrow 0} sG(s)$	where, $K_a = \lim_{s \rightarrow 0} s^2 G(s)$

Steady-state errors for various inputs and Systems Types

Introduction to P-I-D Controllers:

A controller is a device which when introduced in feedback or forward path of system, controls the steady state and transient response as per the requirement. In most of the practical systems, controller output is proportional to error generated. Such systems are called as Proportional to error mechanisms (P type).

Consider such second order system where controller input is error itself and proportional constant is $K=1$ as shown below.



$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

For this system damping ratio is ξ and natural frequency ω_n .

Steady state error, $K_p = \lim_{s \rightarrow 0} G(s)H(s) = \infty$

$$\text{and } K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \frac{\omega_n}{2\xi}$$

If transient response is to be improved, damping ratio must be changed. In general good time response demands,

- (i) Less settling time
- (ii) Less overshoot
- (iii) Less rise time
- (iv) Smallest s.s error

By increase in K_v i.e. increase in system gain, s.s error can be reduced but due to high gain, settling time and peak overshoot increases. This may lead to instability of system.

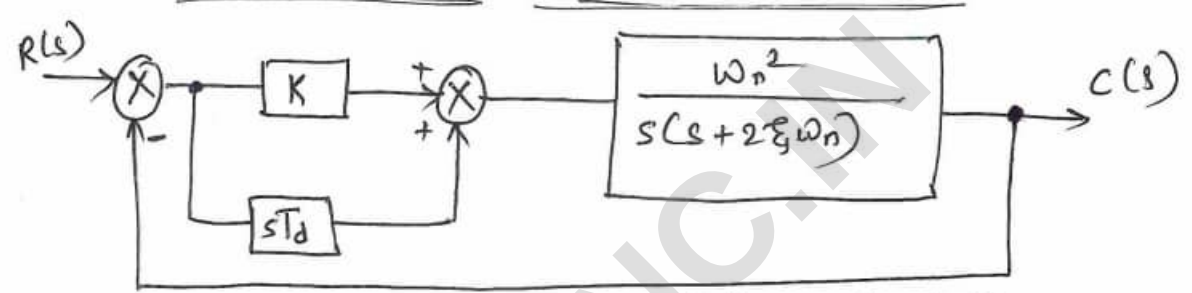
So compromise is made to keep steady state error and overshoot within acceptable limits by providing following different types of controllers.

- (i) PD $\rightarrow$ Proportional + Derivative action
- (ii) PI $\rightarrow$ Proportional + Integral action
- (iii) PID $\rightarrow$ Proportional + Derivative + Integral action

(i) PD Type of Controller:

A controller in the forward path, which changes the controller output corresponding to proportional plus derivative of error signal is called PD controller:

i.e Output of controller = $K e(t) + T_d \frac{d e(t)}{dt}$



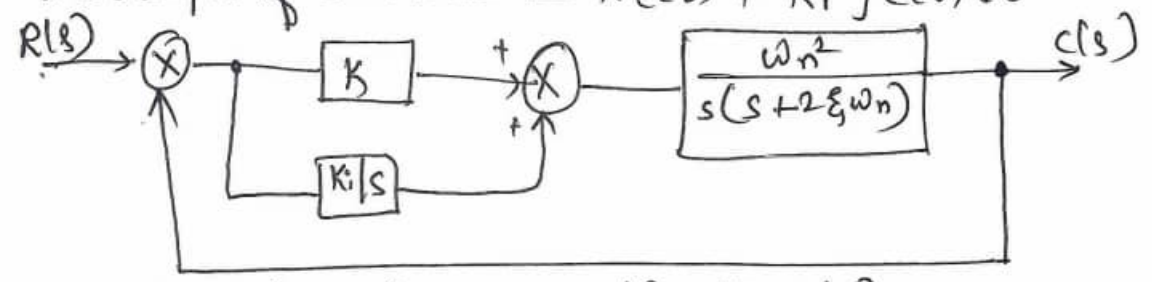
Taking Laplace transform = $K E(s) + s T_d E(s)$
= $E(s) [K + s T_d]$

The transfer function of such controller is $[K + s T_d]$

(ii) PI Type of Controller:

A controller in the forward path, which changes the controller output corresponding to the proportional plus integral of the error signal is called PI controller.

i.e Output of controller = $K e(t) + K_i \int e(t) dt$



Taking Laplace transform = $K E(s) + \frac{K_i}{s} E(s)$
= $E(s) [K + \frac{K_i}{s}]$

The transfer function of such controller is $[K + \frac{K_i}{s}]$

PID Type of Controller:

As PD improves transient and PI improves steady state, combination of two may be used to improve overall time response of the system. This can be realized as shown below.

